

Task register — final-dataset

Written 2026-08-07. The #NN ids are referenced throughout IMPLEMENTATION-PLAN.md , BUILD-DEPENDENCY-GRAPH.md and HANDOFF-FINAL-DATASET.md . They previously lived **only in a session task tool**, so a fresh agent could read #21 in three documents and find no definition anywhere. This file is the definition. **It is the crosswalk between all three numbering schemes** — task id, graph node, and Phase 0 item.

Nothing here has been submitted to AWS. Every job needs a platform submission and a human release.

The schedule these tasks add up to, so nobody re-derives it from a stale figure:

	value
critical path	15.5 h if #28 uses registry rows · 23.5 h via ordinal ranges · 54.5 h if DCLM is not split
job time on the path	11.0 h — jobs and code are now comparable
CPU compute cap	384 vCPU (c7i.8xlarge), MEASURED — not the 128 older text assumes
build floor, 1.0T	9.96 h — $384 \text{ vCPU} \times 72,615 \text{ tok/s/vCPU MEASURED}$; ~78% of cost is the serial Python filter, so encode_batch 's 0.328 M is $4.52\times$ optimistic

Two coupling rules that are easy to miss and change what "done" means:

- **#10 before #29.** Retiring verify --deep exposes GA1 ; unthreaded that is 5.08 h and the path is 24.90 h whether the re-hash is there or not, so **#29 measures as nothing until #10 lands.**
- **#28 is MANDATORY, and its route is a decision.** Un-split, DCLM is **49 h as one child** against a 9.96 h floor. **N registry rows on disjoint subdirectories (~2 h) is the route that works** — the domain_column fan-out does **not** (_domain_of never sees the file entry). **Two of its traps fail silently** — read artifacts/impl-plan/task-28-briefing.md §2.1. **And it must cover FineWeb-Edu (30.1 h) as well as DCLM.**

Wave 0 — the critical-path code

#	task	graph	Phase 0	changes the plan?	est
#28	Split the BIG bundles — MANDATORY, and the ROUTE matters. <code>--shard/--</code> of strides <i>bundles</i> , so DCLM's 410B is one child at 49 h even on a whole 32-vCPU instance against a 9.96 h floor; FineWeb-Edu's 252B is 30.1 h. Use N registry rows on disjoint subdirectories (~2 h → path 15.5 h). ❌ The <code>domain_column</code> fan-out does not work. Fallback is ordinal ranges (12 h → 23.5 h). ⚠️ Two traps fail silently — read <code>artifacts/impl-plan/task-28-briefing.md §2.1</code> first, and add a <code>source_label</code> uniqueness check to <code>load_registry</code> before anything	C3b	3b	YES	~2 h, or 1–2 d
#22	Replace the dedup set with a flat <code>np.uint64</code> pre-pass. The set needs 27.92 GB for DCLM in a 15.03 GB container (§5.2a)	A2a + A2b	4	no	1 d
#21	Wire the FinePhrase id partition into <code>_reader_for</code> . Cannot be retrofitted after tokenization. Ships with the budget correction or not at all	C1	1	YES	~5 lines
#9	Decide <code>SHARD_TOKENS</code> . Code says 25,001,984; the report now agrees. Confirm and stop carrying two values	B6	11	YES	1 h
#24	Rebuild the decon index from raw benchmark fields, not 5-shot renders. Gates FREEZE	B5	12	no	4 h

Wave 0 — parallel, distinct files, no coordination

#	task	graph	Phase 0	est
#23	Pin <code>tokenizers</code> in <code>pyproject.toml</code> . It is imported at <code>corpus_build.py:631</code> and declared nowhere	B1	5	1 line
#10	Thread the Gate A profile checks + raise <code>max_pool_connections</code> . ~100 lines, not the ~20 an earlier draft said. Ship this BEFORE #29 — deleting <code>verify --deep</code> exposes GA1, and unthreaded that is 5.08 h, which makes #29 worth exactly nothing	B3	6	1 d
#29	Declare <code>ChecksumSHA256</code> on the shard upload, then retire <code>verify --deep</code> . <code>corpus_build.py:463</code> calls a plain <code>put</code> and computes the digest one line too late, so S3 never verifies it. A verified PUT is rejected server-side on mismatch (proven live: <code>BadDigest</code> , and no object is created), and <code>CopyObject</code> recomputes on the publish hop — so the 1.49 h re-hash is redundant once the checksum is declared . Keep the code for VENDORED corpora	B7	13	~5 lines
#16	Drop <code>data_provenance_initiative</code> — ships GSM8K in Flan CoT format at 6 repeats. 0.51% of tokens	B4	9	registry edit
—	Fix the boundary-marker prefix guard in <code>corpus_pack.py</code> , and the test asserting the table length is 1	B2	8	~5 lines
—	Record <code>FilterStats</code> in the receipt. Until then no dedup claim is auditable	—	7	~10 lines
#11	Query the live validator timeout. <code>edullm-validator:12</code> 's is recorded nowhere. Read-only	—	10	1 call

Wave 0 — measurements

#	task	graph	est
#26	Measure in-region S3 + HF CDN bandwidth. RUN THIS FIRST — every read estimate borrows ~85 MB/s from an S3 measurement, and one reconciliation implies the CDN is ~8.4 MB/s	M1	10 min
#14	Dolma 3 adult-content prevalence, sampled at random offsets — a prior attempt could not separate signal from HuggingFace preview ordering. Blocking for that source	M2	1 h
—	~~Nemotron-CC-Math's real dolma2 count~~  DONE 2026-08-07 — 134.0B (472,213,218,716 bytes × 0.283686 tok/byte, 1,920 random-offset docs, seed 42; 3 ≈ 83.6B + 4plus ≈ 50.4B). Measured by a teammate with gate access. This closed the M3 node, NOT task #17	M3	done
#17	Re-run Marin's 13-gram contamination scan on Nemotron-CC-Math. A different task from the count above, and still open. The original scan found 11,868 contaminated documents against MATH500 alone — 13.2× the publisher's entire reported removal budget — including verbatim GSM8K test items at Jaccard 1.0. Now unblocked: the gate is accepted and 107+ GiB is already staged in S3, so this runs in-region against staged bytes	—	1 job
—	Mean doc length for 5 unmeasured stage-2 sources. The dolma3 QA source is the one plausibly near the 20-token EOS floor	M4	1 h

Deferrable — do NOT put in the first image

#	task	graph	why deferred
#25	File-shard the VAL bundles. 43% of bytes moved serves 0.39% of tokens. NOT the same item as #28 — this one is a read-volume saving, #28 is a wall-clock prerequisite	C3	∞ slack; contends with #21 on _reader_for
—	Wire bytes_fetched through from corpus_read to the build	C11	instrumentation only

Then, in order

#	task	note
#20	Freeze the full two-stage plan. THE REAL GATE. Its duration is a decision, not a computation	after every "changes the plan" item lands
#6	Ingest. Stage → smoke test → build in waves → publish two datasets → Gate A → verify --deep → promote	
#1	Register reservoir-dolma2 with the platform	independent of this corpus

Downstream of the corpus — training-side, not build-side

#	task
#19	Fix <code>MoELoadBalancingLossGranularity</code> before any training run. Cannot be annealed in — switching at 10% recovers ~55%
#7	Run probe-fitted mixture selection before the real run
#18	Use BPB, not accuracy, for any ablation at this scale — MCF is random below ~400B tokens
#8	Store MTLD as a per-document label at ingest — labels are inside <code>manifest_sha256</code> , so unbackfillable
#13	Settle SuperBPE empirically with the existing byte-matched A/B corpora (~\$600)
#12	Fix the missing EOS in the SuperBPE tokenizer. Affects two already-published corpora, NOT this one — dolma2's <code>eos_token_id</code> is 100257 and its check runs

Closed

#	task	outcome
#5	Measure the real FinePhrase overlap	0.2683 distinct on a complete-column read of 287,000 ids
#15	Interleave sources within shards	Not expressible in data. <code>labels</code> is one dict per shard path and Gate A compares by full dict equality. Trainer-side (#19)
#4	Wire <code>keeps_id</code> into the build path	superseded by #21, which is the same fix scoped to the reader
#27	Fix 23 verified doc inconsistencies	this pass